



BERTweet.BR: a pre-trained language model for tweets in Portuguese

Fernando Carneiro¹ · Daniela Vianna² · Jonnathan Carvalho³ · Alexandre Plastino¹ · Aline Paes¹

Received: 30 January 2024 / Accepted: 19 September 2024 / Published online: 21 December 2024
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Abstract

Recent advancements in neural language models have been primarily centered around English, with limited focus on the more than seven thousand other languages. This includes Portuguese, which, despite being the sixth most spoken language globally, has markedly fewer neural-based linguistic resources available when compared to English. Notably, Portuguese speakers compose one of the most active groups of Twitter users; however, no pre-trained language model in Portuguese tweets is extensively studied in the literature. Besides the language, tweets-based pre-trained models must account for the cultural code, informal linguistic style, code-switching, and the limited number of characters. This manuscript tackles this gap by introducing **BERTweet.BR**, the first public large-scale pre-trained model specific to the Brazilian Portuguese tweets domain. BERTweet.BR has the same architecture of BERTweet_{base}, a BERT-based English-tweets model, and was trained from scratch following the RoBERTa pre-training procedure on a 100-M Portuguese tweets *corpus*. On the sentiment analysis task, experiments show that BERTweet.BR outperforms three multilingual Transformers and BERTimbau, a monolingual general-domain Brazilian Portuguese language model. We release our model in the transformers library aiming at promoting future research in analytical tasks for Portuguese tweets. The *BERTweet.BR* code, experimental results, and related documentation are publicly available on Github.

Keywords Social media · Short noisy texts · Monolingual language model · Sentiment analysis

1 Introduction

The introduction of contextualized embeddings—which address polysemy by allowing distinct embeddings for the same word, depending on the context it appears—opened up a new era for deep learning-based models prone to NLP tasks [19, 39]. Notably, the Transformers [51], further explored in models like BERT [13] and GPT [41], demonstrated that pre-trained multi-layer language models based on attention mechanisms could be easily embedded into a transfer learning strategy to obtain state-of-the-art results in a wide range of downstream tasks, even in scenarios with only a few labeled data. Following BERT, multiple contextualized language models have been trained from heterogeneous and conventional text corpora extracted from sources such as books, Wikipedia, and news sites [24, 43, 58].

While those generic-domain representations have achieved remarkable performance across many tasks with multiple datasets taken from a variety of sources [53, 54], studies have shown that training domain-specific language models can deliver significant gains when dealing with specific textual contexts [17, 26]. BERTweet [34], for

<https://huggingface.co/melll-uff/bertweetbr>.
<https://github.com/MeLLL-UFF/BERTweet.br>.

✉ Jonnathan Carvalho
joncarv@iff.edu.br

Fernando Carneiro
fernandopc@id.uff.br

Daniela Vianna
dvianna@jusbrasil.br

Alexandre Plastino
plastino@ic.uff.br

Aline Paes
alinepaes@ic.uff.br

¹ Institute of Computing, Universidade Federal Fluminense, Niterói, RJ, Brazil

² JusBrasil, Salvador, BA, Brazil

³ Instituto Federal Fluminense, Itaperuna, RJ, Brazil

example, was trained exclusively from tweets in English to capture the informal characteristics present in short texts typical of the Twitter platform. Experiments showed that BERTweet outperforms the baselines RoBERTa_{base} [58] and XLM-R_{base} [11] on three Twitter NLP tasks.

Regarding languages other than English, there is some effort on domain adaptation of large-scale multilingual language models to tweets [3]. However, a question arises whether the performance of monolingual models is better, given the specific particularities of tweets. This way, many recent works have released large-scale language models adapted to the Twitter domain in languages other than English, such as French [16], Indonesian [23], Spanish [15, 20], Arabic [1], and Italian [40].

On the other hand, several thousand other languages remain neglected. Portuguese, for example, is the sixth most spoken language in the world, with 258 million Portuguese speakers [14], and the fifth most used language on the Internet [21]. Brazil has the world's largest population of speakers from Portuguese-speaking countries: approximately 214 million people. Even though only 70% of Brazilians have regular internet access, Brazilians spend more time on the Internet than watching TV. Indeed, Brazil is responsible for 10% of the total time spent on social media globally, positioned in second place, only behind the USA [12]. Along with Spanish, Japanese, and Indonesian-speaking users, Portuguese speakers are among the most active voices on Twitter [18], being the fifth country regarding the number of Twitter users [48]. Even though Twitter's decision to shut down its free API posed challenges for researchers, the study of informal language on social media remains a worthwhile endeavor. The informal language style is not exclusively to Twitter; it is prevalent across many other social media platforms. As a result, the models and insights gained from Twitter data can be directly applied to other social media platforms as well.

These numbers indicate the enormous potential and need for developing language models specifically addressing Twitter-based NLP tasks in Portuguese. Although some previous studies fine-tuned contextualized embeddings for Portuguese tweets [46], the question remains whether a model trained specifically for Portuguese tweets performs better than adapting existing multilingual or formal language models to tweets-based tasks. However, no pre-trained language model from a large-scale corpus of tweets exclusively for Portuguese has been extensively studied in the literature.

In this context, we introduce *BERTweet.BR*, a public large-scale pre-trained model specific to the Brazilian Portuguese tweets domain. *BERTweet.BR* has the same architecture of BERTweet_{base} [34]. Likewise, it was trained from scratch following RoBERTa [58] pre-training

procedure on a *corpus* of approximately 9 GB containing 100 M Portuguese tweets.

To evaluate *BERTweet.BR*, we selected the sentiment analysis task [27], given its broad application scenarios, enabling companies and governments to gain valuable insights into people's attitudes and perceptions, from political opinion [44] to stock market [35]. The evaluation pipelines we employed include a collection of eight manually-annotated datasets, five of which have three classes, while the rest are binary. We compared the performance of *BERTweet.BR* to a broad set of contextualized transformer-based models containing language-specific, multilingual, and Twitter-adapted models. To determine the adoption of *BERTweet.BR* and show its predictive superiority on that task, we conducted two groups of experiments: (i.) adapting the language model that induces contextualized embeddings; and (ii.) training classifiers from ready-to-use or adapted embeddings coming from the language models. Considering these possibilities, we designed experiments to investigate the following research questions.

- **RQ1** *How does BERTweet.BR compare to adapting existing language models to the tweets domain?* We follow the two approaches of Gururangan et al [17] to adapt pre-trained language models and unfold this research question in the following two.
 - **RQ1.1** *What is the predictive performance gain of the pre-trained BERTweet.BR model when compared to adapting existing generic monolingual and multilingual models to the generic domain of tweets?* To answer this question, we adapted the most used Portuguese monolingual model, BERTimbau [46], and the multilingual models mBERT [13] and XLM-R [11] to the same dataset BERTweet.BR was pre-trained following the Domain Adaptative Pre-Training (DAPT) procedure [17].
 - **RQ1.2** *What is the predictive performance gain of the pre-trained BERTweet.BR model when compared to adapting existing generic monolingual and multilingual models to sentiment-prone tweets?* To answer this question, we adapted the models mentioned in RQ1.1 to tweet sentiment datasets following the Task Adaptative Pre-Training (TAPT) procedure [17].
- **RQ2** *What is the difference in the predictive performance of fine-tuning to the downstream tasks the tweets-trained or adapted embeddings compared to using them without fine-tuning?* This research question unfolds in the following two.
 - **RQ2.1** *What is the predictive performance gain of extracting the embeddings from BERTweet.BR,*

when compared to executing this same procedure in the other contextualized models? We followed a feature-based approach to answer this question and trained a logistic regression classifier with embeddings extracted from BERTweet.BR, original and adapted BERTimbau, mBERT, XLM-R. Additionally, we elicited another model trained from multilingual tweets, namely, the XLM-T [3] model.

- **RQ2.2** What is the predictive performance gain of fine-tuning BERTweet.BR compared to executing this same procedure in the other contextualized models, BR to the downstream datasets? To answer this question, we fine-tuned the original and adapted contextualized models mentioned in RQ2.1.

In addition to answering the research questions, we compare BERTweet.BR model's predictive performance to a fastText-based classifier [31] as a baseline result. Experiments showed that our model consistently outperforms mBERT, BERTimbau, XLM-R, and XLM-T in most cases and the static word embeddings from fastText [31] in all the experiments.

In brief, we highlight the main contributions of this work:

- We introduce BERTweet.BR, a large-scale pre-trained language model for Brazilian Portuguese tweets. We empirically show that it outperforms a large set of training strategies and models in sentiment classification using datasets of distinct characteristics and sizes, certifying the effectiveness of a domain-specific language model pre-trained for Portuguese tweets.
- A set of experiments investigating how static and transformer-based word embeddings for Portuguese trained on domains other than tweets perform in the Twitter sentiment analysis task.
- To facilitate future research on Portuguese tweets, we made BERTweet.BR publicly available on Huggingface's model hub.¹ Also, we open-sourced all the code and documentation on Github.²

The rest of the text is organized as follows. Section 2 describes some related studies in the literature. Section 3 outlines the architecture, dataset, and optimization setup adopted to pre-train the BERTweet.BR language model. Section 4 presents the workflow of the experiments carried out in this study. In Sect. 5, we present the experimental results achieved by responding to the research questions introduced in this section. In Sect. 6, we bring a qualitative analysis of the conducted experiments. Finally, in Sects. 7

and 8, we present the conclusions of this study and future research directions, respectively.

2 Related work

Natural language processing has seen rapid advancements in recent years mainly due to the extensive use of transfer learning from deep contextualized pre-trained language models. In *sequential transfer learning*, the source and target tasks are different, and training is performed in two steps: *pre-training* and *adaptation* [42]. The general practice is to pre-train representations on a sizeable unlabeled text corpus (pre-training phase) and then adapt these representations to a supervised target task using labeled data (adaptation phase). To maximize the usefulness of sequential transfer learning, the pre-training task should produce a multipurpose representation of the language that might be useful not for one specific but for several target tasks. Intuitively, one basic approach to pursue such universal representations is to pre-train a language model (LM) on a large general-domain corpus extracted from well-written sources readily available on the Internet.

Following this paradigm, Vaswani et al [51] introduced the Transformer architecture with a novelty self-attention mechanism that dispenses recurrence and convolutions, denoting a great leap forward for NLP tasks with long-term dependencies. After that, influential models have been proposed, including the *auto-regressive* GPT [41] and *auto-encoding* BERT [13], either building upon the decoder or the encoder component of the original Transformers work [51]. GPT implements an *unidirectional* 12-layer *decoder-only* language model pre-trained for 100 epochs on BooksCorpus [57], a dataset of 7,000 books. BERT [13] overcomes the gaps of unidirectionality by introducing a multi-layer *bidirectional* and *encoder-only* Transformer model. The model advanced the state-of-the-art for both sentence-level and token-level NLP tasks after being pre-trained on BooksCorpus (800 M words) and English Wikipedia (2,500 M words) using two unsupervised tasks: the masked language model task (MLM) and next sentence prediction (NSP). One of the relevant variations of BERT is RoBERTa [58], which achieved a significant performance gain by being trained on a corpus considerably larger than BERT [13], discarding the next sentence prediction task (NSP) as well as a longer pre-training phase with larger batches.

Parallel work on multilingual understanding extends these systems to more languages, thus enabling the use of language models beyond English. For example, a multilingual version (mBERT), pre-trained on the 100 largest languages in Wikipedia, was also released along with the original BERT [13]. Despite all the contributions of a

¹ <https://huggingface.co/melll-uff/bertweetbr>.

² <https://github.com/MeLLL-UFF/BERTweet.br>.

multilingual model, authors of BERT acknowledged that the “multilingual model is somewhat worse than a single-language model”. Then, specialized monolingual models have been published, outperforming previous multilingual benchmarks as seen for French [25, 29], Vietnamese [33], Spanish [8], and German [10] among others. For Portuguese, Souza et al [46] replicated BERT’s architecture and pre-training procedures to yield BERTimbau, a language model for Brazilian Portuguese that achieved state-of-the-art performances on three downstream NLP tasks after being pre-trained on brWaC [52], a large and diverse corpus of web pages.

All aforementioned language models were pre-trained from large volumes of conventional text corpora from books, Wikipedia articles, and news sites. However, despite being able to achieve robust performance across different downstream tasks in various languages, when applied to specialized domains, such as biomedical, scientific, or legal texts, the generic-domain representations have been shown to underperform in many target domain tasks, as shown in works like Lee et al [26]; Beltagy et al [4]; Chalkidis et al [9]. As demonstrated by Gururangan et al [17], the more distinct the target and source domains are, the greater the degradation of generic models (source) when used in specific domains (target), and the higher the potential of techniques for model language domain adaptation. This is the case of the microblog Twitter, where users communicate with each other informally, using typical expressions of social networks slang, and many times with lexicon-syntactic errors or adding special tokens such as hashtags, user mentions, and emojis. Therefore, there is an apparent mismatch between the domains of Twitter (target domain) and Wikipedia, books, and news articles (source domain)—traditionally used for pre-training generic-domain language models.

To fill that gap, Nguyen et al [34] proposed BERTweet, a domain-specific language model trained from scratch on English tweets. Studies have demonstrated that BERTweet outperforms baselines such as RoBERTa_{base} [58] and XLM-R_{base} [11] on three NLP tasks on tweets, thus demonstrating the effectiveness of a large-scale language model specially adapted for the specific domain of tweets. Following similar procedures, Polignano et al [40]; Angel González et al [15]; Huertas-Tato et al [20] and Abdelali et al [1] introduced ALBERTo, TWilbert, Bertuit, and QARiB, respectively—pre-trained language models trained from scratch on massive corpora of Italian, Spanish (two models), and Arabic tweets. Later, XLM-T [3] was proposed following the DAPT procedure [2, 17] to adapt XLM-R [11] to the Twitter domain, creating a multilingual model from a corpus containing 198 M tweets written in the 30 most represented languages in Twitter. Guo et al [16] continued pre-training the generic-domain model

CamemBERT [29] to provide BERTweetFR, a domain-specific model for French tweets. Koto et al [23] trained five different versions of the IndoBERTweet model to compare the following approaches: pre-training from scratch based in IndoBERT [22] model along with four other adapted models resulting from the domain-adaptive pre-training procedure with distinct vocabulary adaptation strategies. The study reveals that it is feasible to adapt an off-the-shelf pre-trained model very efficiently and obtain better average performance than training from scratch.

Portuguese, one of the most active languages on Twitter, has been remarkably neglected by this tendency to pre-train massive language-specific LM for tweets. That is where the scope of this work applies. We propose a replication of BERTweet aiming to advance the research of NLP in Portuguese on top of a language model capable of accurately inheriting leanings, nuances, and biases from Portuguese tweets, which can later be usefully applied to any downstream task in Twitter and informal scenarios.

3 Building BERTweet.BR

This section outlines the architecture, dataset, and optimization setup adopted to pre-train the *BERTweet.BR* language model.

3.1 Architecture

BERTweet.BR follows the same architecture and pre-training procedure as the BERTweet model [34], replicating RoBERTa and pre-trains the model based on the masked language model objective only. We employed a multi-layer bidirectional Transformer architecture using the same configuration as the base version of BERT [13] with 12 layers, 768 hidden dimensions, and 12 attention heads, adding up to 110 M parameters.

3.2 Tokenizer

We adapted the normalization step of the original BERTweet tokenizer (BERTweetTokenizer) to deal with demojizer for Portuguese. The *BERTweet.BR* tokenizer was trained to have a 64K token vocabulary and use fastBPE [45] to segment words into subword units. We set the maximum sequence length of the tokenizer to 128.

3.3 Pre-training dataset

We downloaded tweets from the collection grabbed by the Archive Team,³ containing tweets streamed from the

³ <https://archive.org/details/twitterstream..>

general Twitter stream from 2004 to 2020. We filtered Portuguese tweets by setting the field *lang* to *pt* (“*lang*” = “*pt*”). We tokenized sentences using the *TweetTokenizer* class from the NLTK library [5] and used the *demojize* method of the emoji library⁴ to translate emotion icons into text strings in Portuguese. We also converted user mentions and web or URL links into the special tokens @USER and HTTPURL, respectively. We then obtained a corpus of 100 M unique tweets split into 90/10 percent proportion for training and test sets, respectively. The resulting dataset is approximately 9 GB large.

3.4 Pre-training

BERTweet.BR was pre-trained on the Masked Language Modeling (MLM) task. In any given input sequence, 15% of the tokens were chosen for possible replacement so that the model was subsequently trained to predict tokens replaced by <MASK> using cross-entropy loss. We leveraged the RoBERTa implementation of the Transformers library [55] and followed the language model training script *run_mlm.py*⁵ with the PyTorch [37] distributed package in a Linux CentOS 7 server.

We optimized the model using Adam with a batch size of 96 across 4 GPUs (NVIDIA Tesla V100-SXM2-32GB) and a peak learning rate of 0.0001. We pre-trained *BERTweet.BR* for 30 epochs in about three weeks (the first 50K training steps were used for warming up the learning rate). The model was evaluated every 50K steps during pre-training as illustrated in Fig. 1. We then named *BERTweet.BR* the very last checkpoint as that was the best-performing version after approximately 7 M training steps.

4 Experiments design

The experiments conducted in this section aim at answering the research questions introduced in Sect. 1:

- **RQ1** How does *BERTweet.BR* compare to adapting existing language models to the tweets domain?
 - **RQ1.1** What is the predictive performance gain of the pre-trained *BERTweet.BR* model when compared to adapting existing generic monolingual and multilingual models to the generic domain of tweets?
 - **RQ1.2** What is the predictive performance gain of the pre-trained *BERTweet.BR* model when

⁴ <https://pypi.org/project/emoji/>.

⁵ https://github.com/huggingface/transformers/blob/main/examples/pytorch/language-modeling/run_mlm.py.

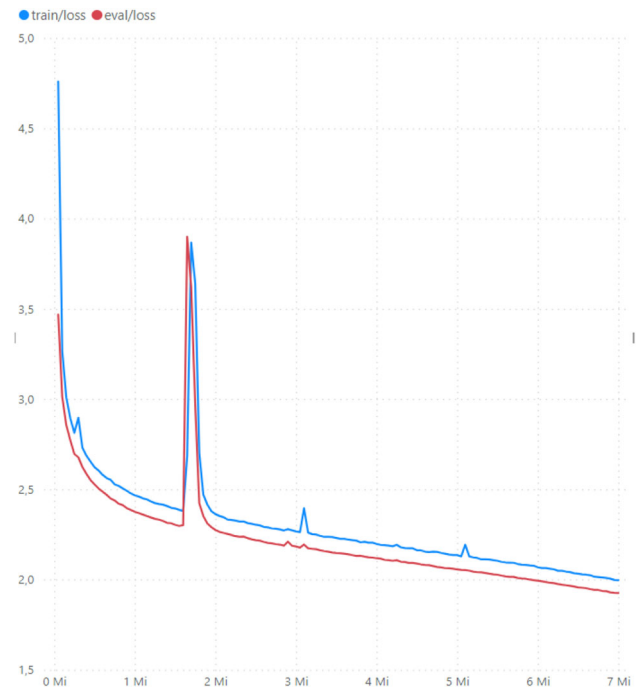


Fig. 1 Training and evaluation loss progress of BERTweet.BR model, trained from scratch for 30 epochs

compared to adapting existing generic monolingual and multilingual models to sentiment-prone tweets?

- **RQ2** What is the difference in the predictive performance of fine-tuning to the downstream tasks the tweets-trained or adapted embeddings compared to using them without fine-tuning?
 - **RQ2.1** What is the predictive performance gain of extracting the embeddings from BERTweet.BR when compared to executing this same procedure in the other contextualized models?
 - **RQ2.2** What is the predictive performance gain of fine-tuning BERTweet.BR to the downstream datasets when compared to executing this same procedure in the other contextualized models?

To this end, we benchmark the performance of *BERTweet.BR* in the task of sentiment analysis in tweets using a set of different pipelines. The evaluation includes static to contextualized embeddings, ranging from language specific to multilingual models and Twitter-specialized options. The former relies on fastText. The latter leverages transformer-based architectures with a comprehensive list of four models: BERTimbau, mBERT (multilingual version of BERT), XLM-R (multilingual version of RoBERTa), and XLM-T, an adaptation of XLM-R for the Twitter domain.

When applicable, strategies for applying pre-trained language representations to downstream tasks encompass a

Table 1 Summary of the experiment baselines of each model language and domain adaptation procedures explored in this work

	Original	TAPT	DAPT
FastText	✓		
MBERT	✓	✓	✓
XLM-R	✓	✓	✓
XLM-T	✓		
BERTimbau	✓	✓	✓
BERTweet.BR	✓		

feature-based and a *fine-tuning* procedure. Therefore, for all transformer-based models, we conduct both strategies, while for the pre-trained word vector fastText, to which *fine-tuning* is not applicable, we only employed the standard *feature-based* strategy.

We compare *BERTweet.BR* with up to three different benchmark of each model as shown in Table 1. First and foremost, we conducted the experiments on top of the *off-the-shelf* publicly available pre-trained weights for all the benchmark language models. These are the original pre-trained versions released by their authors, called here as *original checkpoints* (column *Original*).

Then, we employ the continuous pre-training strategy to account for adapting the original checkpoints to the tweets domain as per the research question *RQ1*.

To this end, we adopted the *domain-adaptive pre-training* (DAPT) and *task-adaptive pre-training* (TAPT) procedures of [17]. Then, regarding *RQ2*, the resulting models were used to perform the same set of experiments of sentiment analysis using again the *feature-based* and *fine-tuning* strategies, now over these two additional variations of language models (DAPT and TAPT). In summary, we compare *BERTweet.BR* to a total of 11 different versions of language models (Table 1). As we apply both *feature-based* and *fine-tuning* approaches to contextualized LM and only the first of the methods to fastText, we reach 23 experiments in this study.

4.1 Sentiment analysis experiment

Sentiment analysis is a supervised sequence classification task in which we inspect a given text and identify the prevailing opinion within it, typically to determine a writer's attitude as *positive*, *negative*, or *neutral*. In this work, given a specific tweet, the goal is to determine whether it reveals a positive or negative opinion (binary mode) or expresses a neutral, positive, or negative message (multi-class mode).

Sentiment analysis is a crucial task in natural language processing that enables governments, organizations, and

other entities to gain valuable insights into people's attitudes and perceptions toward a wide range of topics. The immense significance this downstream task holds makes it a natural choice when evaluating the benefits of language models.

The application of sentiment analysis has far-reaching implications across various domains, including marketing, politics, and healthcare. For instance, sentiment analysis can assist marketers in understanding consumer preferences and feedback, helping them to tailor their campaigns and improve customer engagement. Similarly, in the political domain, sentiment analysis can be leveraged to gauge public opinion on governmental policies and actions, and to monitor the spread of propaganda and misinformation. Also, during crises such as pandemics or natural disasters, sentiment analysis can provide real-time insights into the public's sentiments, allowing policymakers to gauge public reactions and implement measures accordingly.

Datasets We retrieved a plural set of eight human-annotated datasets from various domains for sentiment analysis. As described in Table 2, they are also varied in size and number of classes, with three being binary datasets containing *negative* and *positive* classes. In contrast, five others are ternary, having the additional *neutral* label. For each dataset, Table 2 shows the total number of tweets (column *tweets*) and the number and percentage of rows labeled with each of the three classes (columns *positive*, *negative*, and *neutral*). Datasets are grouped by the number of classes (rows *binary* and *multiclass*) and ordered by the number of tweets. Also, it is provided the number and percentage of duplicated tweets (retweets – column *duplicated*) along with the minimum, maximum and average length of tweets of each dataset (columns *min*, *max*, and *avg*, respectively). Length is computed before normalization by splitting tweets in white space. These labeled datasets can be downloaded from <https://bityli.com/RvhFax..> We also made the collection available in the transformers datasets library (Table 2).⁶

Evaluation As proposed in research question *RQ2*, we apply *feature-based* and *fine-tuning* approaches to sentiment analysis to assess the reliability of *BERTweet.BR* language model. We evaluate the performance of models on each dataset following a 10-fold stratified cross-validation strategy in which the sample distribution for each class is preserved in all folds. The results are expressed by the average weighted F1-score across the ten folds on the validation set. Datasets are shuffled before splitting. We ensure experiments across models are comparable even in the fold level as a *seed* is set to control the randomness.

⁶ <https://huggingface.co/melll-uff/>.

Table 2 Target datasets grouped by the number of classes and ordered by the number of tweets. Length is computed before normalization by splitting tweets in white space

		Tweets	Positive		Negative		Neutral		Duplicated		Min	Max	Avg
Binary	Covidbr	600	300	50.0%	300	50.0%	–	–	2	0.3%	5	53	28.0
	Sentbr	7769	4773	61.4%	2996	38.6%	–	–	9	0.1%	1	53	11.9
	Fiat	8827	4437	50.3%	4390	49.7%	–	–	0	0.0%	1	58	16.7
		17196	9510	55.3%	7686	44.7%	–	–	11	0.1%	1	58	
multiclass	Narrpt	772	297	38.5%	213	27.6%	262	33.9%	8	1.0%	2	31	14.05
	Mining	2018	166	8.2%	1299	64.4%	553	27.4%	59	2.9%	2	31	14.81
	Compbr	2281	197	8.6%	407	17.8%	1677	73.5%	177	7.8%	1	47	16.44
	Tweemg	8199	3300	40.2%	2446	29.8%	2453	29.9%	2424	29.6%	1	32	16.14
	Unilex	12665	3715	29.3%	4197	33.1%	4753	37.5%	0	0.0%	1	62	14.30
		25935	7675	29.6%	8562	33.0%	9698	37.4%	2668	10.3%	1	62	
		43131	17185	39.84%	16248	37.68%	9698	22.48%	2668	6.18%	1	62	

The values in bold represent the total for each column

Feature-based Approach In the feature-based approach, we use pre-trained representations as input features for a downstream task without changing the original language models. We pass sentences through the model that outputs embeddings which are then used to fit a classifier. In this work, we adopted the logistic regression (LR) classification algorithm due to its simplicity, efficiency, and interpretability. We used the LR implementation from scikit-learn [38].⁷

Static embeddings from fastText [31] are obtained for whole sentences after normalization. Given a *tweet*, we get a single vector representation with *get_sentence_vector* method on top of the Portuguese pre-trained word vectors⁸ shared by the fastText team with the dimension of 300.

When applied to transformer-based models, the *feature-based* approach is comprised of extracting the contextualized embeddings from one or more layers without fine-tuning any parameters. In this work, we sum the outputs of the last four hidden layers as described by Devlin et al [13] when applying BERT with the *feature-based* approach. We pass the normalized *tweets* as inputs and focus only on the first position of the *hidden_states* outputs generated by the pre-trained models, which is the output for the special token used as the aggregate sequence for sentence classification tasks. The resulting contextualized embeddings are a vector of the same size as the pre-trained model's *hidden_size*. All transformer-based models used in this work have *hidden_size* of 768, including *BERTweet.BR*.

The experiments described in this section are aimed at inquiries posed in *RQ2.1*.

⁷ <https://scikit-learn.org/>

⁸ <https://dl.fbaipublicfiles.com/fasttext/vectors-crawl/cc.pt.300.bin.gz>.

Fine-tuning Approach Following a fine-tuning approach is the standard pipeline when employing transformer-based models into target downstream tasks. Fine-tuning is the training done after a language model has been pre-trained, and, as opposed to feature-based, it involves adjusting the pre-trained model's weights on a specific task. Each downstream task has its appropriate fine-tuned models with a task-specific head but initialized with the same pre-trained parameters of language models.

Regarding *RQ2.2*, we employ the framework of transformers library [55] to fine-tune our *BERTweet.BR* and other benchmarking models for the downstream task with each of the eight datasets. To this end, for each dataset and each model, we instantiate a task-specific architecture formed by the original transformer model initialized with the pre-trained weights and an additional untrained sequence classification head on top. Then, using the labeled data from downstream datasets shown in Table 2, all parameters of these combined models are adjusted for sentiment analysis for three epochs. We use *AdamW* [28] with a fixed learning rate of $5e-5$ and a batch size of 32 tweets, reporting results of the last model checkpoint on the validation set.

4.2 Domain adaptation

In order to tackle the issues raised in research question *RQ1*, we also conduct the experiments described in Sect. 4.1 on top of adjusted models, resulting from a second phase of the pre-training procedure from their original checkpoints (continued pre-training). The objective of these experiments is to adapt the three originally general-domain language models—mBERT, XLM-R, and BERTimbau—to the domain of tweets, the object of this

study, and then also compare our *BERTweet.BR* to these additional language models fine-tuned to tweets in Portuguese. To this end, we follow both *domain-adaptive pre-training* (DAPT) and *task-adaptive pre-training* (TAPT) procedures of [17]. In this sense, we seek to verify if only employing a lower-resource pre-training approach through domain adaptation of existing models would produce better results than pre-training a new language model from scratch.

Domain adaptive pre-training (DAPT) The procedure we follow for *domain-adaptive pre-training* (DAPT) is straightforward—we continue pre-training the three target language models on the same unlabeled corpus of tweets we pre-trained *BERTweet.BR*. We also use the same training script *run_mlm.py*⁹ and set of hyperparameters, running on the same hardware configuration, but this time for three epochs. We keep the original tokenizers and their vocabularies. This second phase of pre-training results in three domain-adapted LMs, which are then used in the benchmark experiments of Sect. 4.1 to answer *RQ1.1*. In this work, we refer to these models as <original-model-name>-DAPT as shown in Fig. 2.

Task-adaptive pre-training (TAPT) refers to pre-training an LM on an unlabeled training set that was initially curated for a supervised given task. TAPT is much less expensive because it uses a far smaller corpus but can still produce competitive results for domain adaptation [17].

Instead of continuing pre-training LMs on each of the eight datasets individually, our approach to TAPT considers the entire collection as one augmented dataset. It employs a *LOO* (*Leave One dataset Out*) strategy for training. Specifically, we take each dataset once as the target dataset while the tweets from the remaining seven datasets are combined to tune the language model. Next, in order to answer *RQ1.2*, the resulting model is used to evaluate the performance in the target dataset on the experiments described in Sect. 4.1. Therefore, the procedure repeats eight times for each language model candidate for TAPT: mBERT, XLM-R, and BERTimbau. Note that for the pre-training phase, which is a self-supervised task, we omit the labels of the combined dataset. In contrast, labels are preserved for the supervised sentiment analysis task on the target dataset. To this end, we employ the transformers library [55] and customize the training script *run_mlm_no_trainer.py*.¹⁰ with its default hyperparameter values to continue pre-training models using a masked language modeling loss for 20 epochs. Similarly to DAPT, we also keep the original tokenizers and their

vocabularies. Finally, the last checkpoint produced is then used in the benchmark experiments. In this work, we refer to these models as <original-model-name>-TAPT as shown in Fig. 2.

5 Experimental results

Tables 3, 4, and 5 present results for our *BERTweet.BR* as well as model baselines regarding both off-the-shelf checkpoints and domain-adapted language models resulting from *task-adaptive pre-training* (TAPT) and *domain-adaptive pre-training* (DAPT), respectively. *BERTweet.BR* reported the absolute best F1-score on 36 out of the 48 experiments while matching the previous best performances on two scenarios. Our new model, trained from scratch, achieved the best scores in approximately 80% of all experiments. Taking into consideration the overall results by dataset as shown in Table 6, the *fine-tuning* approach applying *BERTweet.BR* dominates the best scores, yielding the highest f1-scores for six of the eight datasets. Analyzing the results per method, we observe that *BERTweet.BR* registered the best scores in half of the datasets regarding the *feature-based* approach, while for *fine-tuning* our model achieved the highest performances in 75% of the cases. This overview reinforces the predictive effectiveness of *BERTweet.BR* and indicates a positive response to the inquiries of both research questions *RQ1* and *RQ2*.

With respect to *RQ1.1*, Table 5 shows that *BERTweet.BR* outperforms DAPT-adapted models in about 70% of the experiments (11 out of 16). When compared to TAPT models, responding to *RQ1.2*, our model goes beyond, demonstrating gains in over 80% (13 out of 16) of the scenarios, as reported in Table 4. Regarding *RQ2.1*, we isolate the results of the *feature-based* approach from Tables 3, 4 and 5 to highlight that *BERTweet.BR* achieved the highest scores in 18 out of the 24 cases (75%). Now, shifting the focus to the *fine-tuning* approach and responding to *RQ2.2*, we highlight that *BERTweet.BR* obtained the best predictive performance in 20 out of the 24 experiments (83%).

Conversely, the results indicate that *BERTweet.BR* did not perform so well when applied to the *opcovidbr* dataset. Although our model yielded competitive results across all datasets, it failed to obtain the highest score in any of the experiments with *opcovidbr* dataset, with the sole exception of the fine-tuning after DAPT (Table 5). However, this only victory of our model in *opcovidbr* was due to an atypical decrease in performance of BERTimbau in this very specific scenario. One probable explanation for this suboptimal performance of *BERTweet.BR* in *opcovidbr* is that the corpus used to pre-train our model only includes

⁹ https://github.com/huggingface/transformers/blob/main/examples/pytorch/language-modeling/run_mlm.py.

¹⁰ https://github.com/huggingface/transformers/blob/main/examples/pytorch/language-modeling/run_mlm_no_trainer.py.

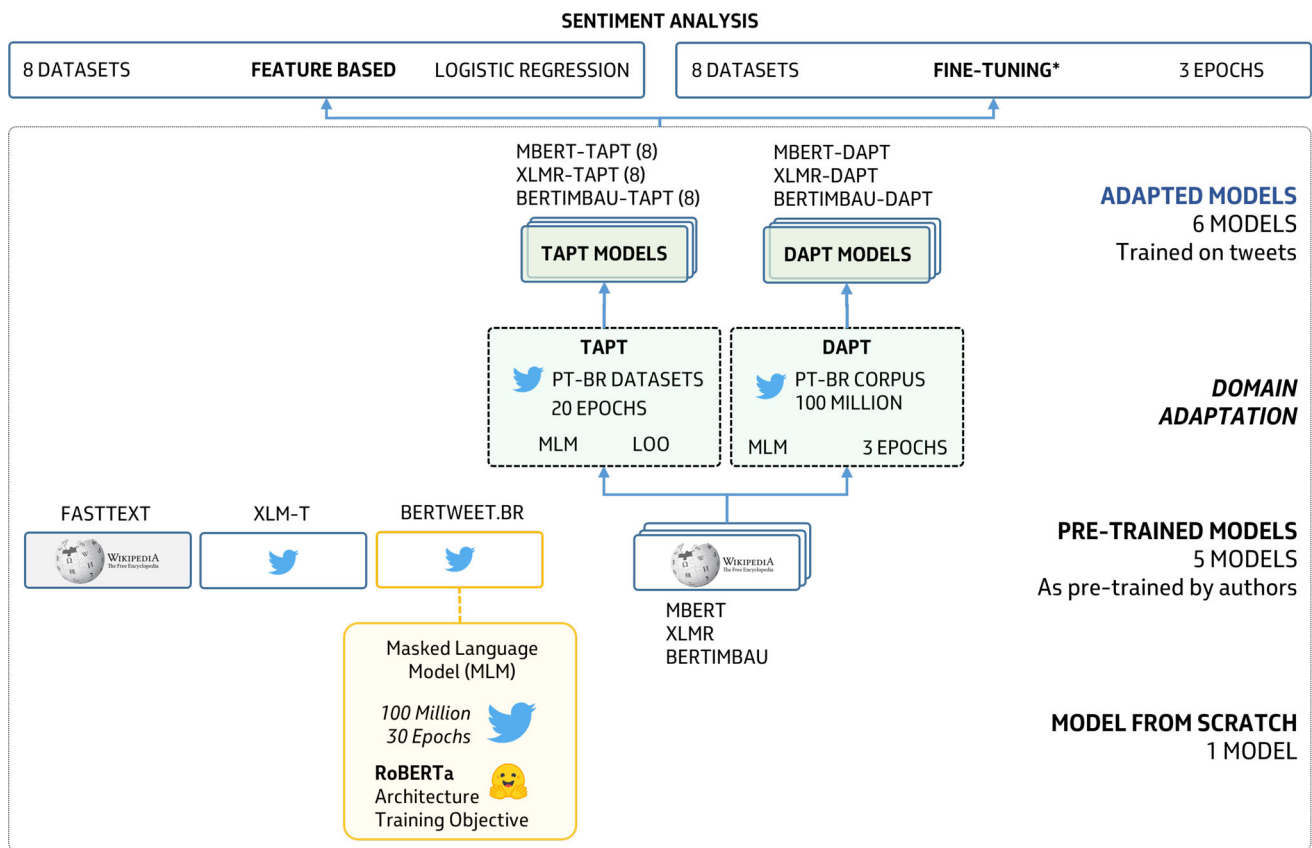


Fig. 2 The summary of experiments. *BERTweet.BR* was pre-trained from scratch on the Masked Language Modeling task using RoBERTa architecture over 100 million tweets for 30 epochs. We then benchmark the performance of *BERTweet.BR* in the task of sentiment analysis in tweets. We use a total of eight datasets and employ both *feature-based* and *fine-tuning* approaches on top of the models. The evaluation includes static to contextualized embeddings, ranging from language specific to multilingual models and Twitter-specialized

options. The former relies on fastText while the latter leverages transformer-based architectures with a comprehensive list of four models: BERTimbau, mBERT, XLM-R, and XLM-T. We also conduct these experiments on top of adjusted models, resulting from a second phase of pre-training procedure from their original checkpoints (domain adaptation) following *TAPT* and *DAPT* procedures of Gururangan et al. [17]

tweets from 2004 to 2020. This may have resulted in a lack of knowledge for this particular domain related to the COVID-19 pandemic, as the debate on the COVID-19 crisis has been mostly made after 2020.

It is worth noting that regardless of not following state-of-the-art architectures based on Transformers [51], the *feature-based* approach with fastText [31] achieved competitive scores and even delivered performances somewhat better than the multilingual models mBERT [13] and XLM-R [11]. In this sense, we can highlight and suggest that in scenarios of low computational resources and restrictions of time, it is reasonable adopting the fastText model in a sentence classification pipeline and still get adequate accuracy in downstream tasks. Therefore, in scenarios where there is a lack of language-specific pre-trained transformer models, using pre-trained static word embeddings can still be a good choice.

Disregarding our *BERTweet.BR*, Portuguese-native *BERTimbau* and Twitter-native multilingual *XLM-T* ranked

the highest among the best-performing models, followed by the generic multilingual *XLM-R* and mBERT. In this setting, without *BERTweet.BR*, *BERTimbau* and *XLM-T* accounted for 75% of the best scores in the experiments. In particular, *BERTimbau* achieved the best results when the *fine-tuning* method was applied, whereas *XLM-T* performed better in most cases when a *feature-based* approach was applied. This is shown by the rightmost column *wins** of Tables 3, 4, and 5 and is consistent with previous conclusions that off-the-shelf general-domain pre-trained models are suboptimal for domain-specific tasks.

In addition, we grouped the results by approach, consolidating the highest scores for each method and dataset, as shown in Table 6. Concerning the two existing strategies for applying language models to downstream tasks, *feature-based* and *fine-tuning*, we observed that *fine-tuning* is indeed the best approach when working with transformer-based models as it outperformed *feature-based* in all scenarios in this study, yielding a 5% boost on average and up

Table 3 The weighted f1-score in the test set for each of the eight datasets for sentiment analysis is on top of the original checkpoints of the models. The models are evaluated in a stratified 10-fold strategy using both *feature-based (fb)* and *fine-tuning (ft)* approaches. In bold, the best performance recorded for each dataset is separated by approach. For each group, in italic, the best performance previous to *BERTweet.BR* is also reported. In parentheses, the percentage gain (or loss) of *BERTweet.BR* results compared to the previous best results. The rightmost columns (wins and wins*) indicate the number of datasets each language model got the best f1-score, respectively, in the scenarios where our *BERTweet.BR* model is considered or not. Note that the fine-tuning approach gives the best performance for all datasets

Original Checkpoint	Feature-based	Fasttext	Covidbr	Sentbr	Fiat	Narrpt	Mining	Compbr	Tweemg	Unilex	Wins	Wins*
<i>feature-based</i>	Fasttext	0.770	0.828	0.815	0.776	0.776	0.776	0.781	0.945	0.661	0	0
	Bertimbau	0.805	0.842	0.826	0.804	0.804	0.784	0.821	0.954	0.668	2	2
	Mbert	0.716	0.738	0.774	0.651	0.706	0.706	0.752	0.927	0.602	0	0
	Xlmr	0.771	0.803	0.807	0.738	0.777	0.777	0.799	0.942	0.649	0	0
	Xlmt	0.768	0.851	0.826	0.838	0.802	0.802	0.834	0.949	0.672	0	6
	Previous	0.805	0.851	0.826	0.838	0.802	0.802	0.834	0.954	0.672	—	—
	BERTweet.BR	0.778 (−3.4)	0.881 (3.5)	0.827 (0.1)	0.843 (0.5)	0.824 (2.7)	0.848 (1.7)	0.855	0.954 (0.0)	0.679 (1.1)	7	—
	Bertimbau	0.864	0.877	0.891	0.855	0.828	0.855	0.873	0.972	0.732	1	5
	Mbert	0.786	0.841	0.878	0.688	0.759	0.773	0.773	0.972	0.703	0	0
	Xlmr	0.768	0.825	0.862	0.742	0.732	0.742	0.742	0.967	0.715	0	0
<i>fine-tuning</i>	Xlmt	0.806	0.882	0.888	0.873	0.829	0.829	0.842	0.971	0.723	0	3
	Previous	0.864	0.882	0.891	0.873	0.829	0.829	0.855	0.972	0.732	—	—
	BERTweet.BR	0.828 (−4.2)	0.902 (2.2)	0.892 (0.1)	0.909 (4.2)	0.857 (3.4)	0.874 (2.2)	0.874 (2.2)	0.973 (0.1)	0.733 (0.2)	7	—
	Bertimbau	Bertimbau	Bertweetbr	Bertweetbr	Bertweetbr	Bertweetbr	Bertweetbr	Bertweetbr	Bertweetbr	Bertweetbr	—	—
	Summary	0.864	0.902	0.892	0.909	0.857	0.874	0.874	0.973	0.733	—	—

Table 4 The weighted f1-score in the test set for each of the eight datasets for sentiment analysis on top of adapted models resulting from *task-adaptive pre-trained (TAPT)* procedure. The models are evaluated in a stratified 10-fold strategy using both *feature-based (fb)* and *fine-tuning (ft)* approaches. In bold, the best performance recorded for each dataset is separated by approach. For each group, in italic, the best performance previous to *BERTweet.BR* is also reported. In parentheses, the percentage gain (or loss) of *BERTweet.BR* results compared to the previous best results. The ↓ indicates the cases where the TAPT models performed worse than the original checkpoints. The rightmost columns (wins and wins*) indicate the number of datasets that each language model got the best f1-score; respectively, in the scenarios where our *BERTweet.BR* model is considered or not. Note that the fine-tuning approach gives the best performance for all datasets

TAPT	Feature-based	Bertimbau	Covidbr	Sentbr	Fiat	Narrpt	Mining	Compbr	Tweemg	Unilex	Wins	Wins*
TAPT	Bertimbau	↓ 0.791	↓ 0.824	↓ 0.825	↓ 0.797	0.789	0.826	0.826	↓ 0.950	↓ 0.654	0	0
	Mbert	0.743	0.767	0.792	0.690	0.721	0.768	0.768	0.932	0.634	0	0
	Xlmr	0.805	0.819	0.825	0.787	0.787	0.814	0.814	0.952	0.657	1	2
	Xlmt	0.768	0.851	0.826	0.838	0.802	0.834	0.834	0.949	0.672	0	6
	Previous	0.805	<i>0.851</i>	<i>0.826</i>	<i>0.838</i>	<i>0.802</i>	<i>0.834</i>	<i>0.834</i>	↓ 0.952	<i>0.672</i>	–	–
	BERTweet.BR	0.778 (–3.4)	0.881 (3.5)	0.827 (0.1)	0.843 (0.5)	0.824 (4.5)	0.848 (1.7)	0.848 (1.7)	0.954 (0.2)	0.679 (1.1)	7	7
Fine-tuning	Bertimbau	↓ 0.853	↓ 0.871	↓ 0.889	0.872	0.834	0.856	0.856	0.974	↓ 0.728	2	6
	Mbert	↓ 0.781	0.843	0.885	0.763	0.801	0.815	0.815	↓ 0.969	↓ 0.702	0	0
	Xlmr	0.799	0.856	0.884	0.786	0.792	0.810	0.810	0.971	↓ 0.698	0	0
	Xlmt	0.806	0.882	0.888	0.873	0.829	0.842	0.842	0.971	0.723	0	2
	Previous	↓ 0.853	0.882	↓ 0.889	<i>0.873</i>	<i>0.834</i>	0.856	0.856	0.974	↓ 0.728	–	–
	BERTweet.BR	0.828 (–2.9)	0.902 (2.2)	0.892 (0.4)	0.909 (4.2)	0.857 (2.7)	0.874 (2.1)	0.874 (2.1)	0.973 (–0.1)	0.733 (0.7)	6	–
summary	Bertimbau	Bertimbau	Bertweetbr	Bertweetbr	Bertweetbr	Bertweetbr	Bertweetbr	Bertweetbr	Bertimbau	Bertweetbr	–	–
	0.853	0.902	0.892	0.909	0.857	0.874	0.974	0.733	–	–	–	–

Table 5 Weighted f1-score in the test set for each of the eight datasets for sentiment analysis on top of adapted models resulting from *domain-adaptive pre-trained (DAPT)* procedure. The models are evaluated in a stratified 10-fold strategy using both *feature-based (fb)* and *fine-tuning (ft)* approaches. In bold, the best performance recorded for each dataset is separated by approach. For each group, in italic, the best performance previous to *BERTweet.BR* is reported. In parentheses, the percentage gain (or loss) of *BERTweet.BR* results compared to the previous best results. The ↓ indicates the cases where the DAPT models performed worse than the original checkpoints. The rightmost columns (wins and wins*) indicate the number of datasets each language model got the best F1-score; respectively, in the scenarios where our *BERTweet.BR* model is present or not. Note that the fine-tuning approach gives the best performance for all datasets

DAPT	Feature-based	Covidbr	Sentbr	Fiat	Narrpt	Mining	Compbr	Tweemg	Unilex	Wins	Wins*
<i>feature-based</i>	Bertimbau	↓ 0.781	0.863	0.830	0.840	0.816	0.834	0.955	0.674	2	3
	Mbert	0.773	0.861	0.821	0.848	0.809	0.830	0.956	0.673	0	0
	Xlmr	↓ 0.766	0.867	0.824	0.861	0.821	0.831	0.961	0.675	2	5
	Xlmt	0.768	0.851	0.826	0.838	0.802	0.834	0.949	0.672	0	0
	Previous	↓ 0.781	0.867	0.830	0.861	0.821	0.834	0.961	0.675	–	–
	BERTweet.BR	0.778 (–0.4)	0.881 (1.7)	0.827 (–0.4)	0.843 (–2.2)	0.824 (0.4)	0.848 (1.7)	0.954 (–0.6)	0.679 (0.7)	4	–
	Bertimbau	↓ 0.824	0.890	↓ 0.889	0.887	0.852	0.864	0.972	0.735	1	2
	Mbert	0.826	0.885	0.888	0.882	0.848	0.866	0.973	0.734	1	2
	Xlmr	0.823	0.894	0.890	0.889	0.849	0.869	0.972	0.731	0	4
	Xlmt	0.806	0.882	0.888	0.873	0.829	0.842	0.971	0.723	0	0
<i>fine-tuning</i>	Previous	↓ 0.826	0.894	↓ 0.890	0.889	0.852	0.869	0.973	0.735	–	–
	BERTweet.BR	0.828 (0.2)	0.902 (0.8)	0.892 (0.3)	0.909 (2.2)	0.857 (0.5)	0.874 (0.5)	0.973 (0.0)	0.733 _(–0.3)	7	–
	Bertimbau	Bertweetbr	Bertweetbr	Bertweetbr	Bertweetbr	Bertweetbr	Bertweetbr	Bertweetbr	Bertimbau	–	–
	Summary	0.828	0.902	0.892	0.909	0.857	0.874	0.974	0.733	–	–

Table 6 Summary of the best results by dataset and approaches feature-based and fine-tuning. The scores are provided as weighted f1-score in the test set for each of the eight datasets for sentiment

analysis. Note that the fine-tuning approach gives the best performance for all datasets. Our *BERTweet.BR* provides 6 out of 8 best scores

	Covidbr	Sentbr	Fiat	Narrpt	Mining	Compbr	Tweemg	Unilex
Feature-based	bertimbau	bertweetbr	bertimbau-dapt	bertimbau-dapt	bertweetbr	bertweetbr	xlmr-dapt	Bertweetbr
	0.805	0.881	0.830	0.861	0.824	0.848	0.961	0.679
Fine-tuning	Bertimbau	Bertweetbr	Bertweetbr	Bertweetbr	Bertweetbr	Bertweetbr	Bertimbau-tapt	Bertweetbr
								Mbert-dapt
								Bertimbau-dapt
	0.864	0.902	0.892	0.909	0.857	0.874	0.974	0.735
	Bertimbau	Bertweetbr	Bertweetbr	Bertweetbr	Bertweetbr	Bertweetbr	Bertimbau-tapt	Bertweetbr
								Mbert-dapt
								Bertimbau-dapt
Best	0.864	0.902	0.892	0.909	0.857	0.874	0.974	0.735

The values in bold represent the best results

to 9% gain over *feature-based*, as reported in *DAPT* for the *Unilex* dataset. Regarding the *feature-based* approach in the context of *RQ2.1*, *BERTweet.BR* achieved the highest scores in 50% of the cases, more than all the other benchmark models. Concerning the *fine-tuning* approach under *RQ2.2*, our model achieved the top performance for six out of the eight datasets in this study.

Regarding the domain adaptation strategy, the results show that TAPT and DAPT both led to performance gains as originally proposed by Gururangan et al [17]. Specifically, the average enhancements observed for TAPT and DAPT over the original checkpoints amounted to 1.8% and 6.3%, respectively. However, these procedures do not always provide benefits. For example, as indicated in Table 4, when applied to Portuguese-native BERTimbau and multilingual mBERT models, *TAPT* resulted in a loss of performance or null gains in a few cases. We found that *DAPT* consistently boosts the performance of the original models, yielding a gain of up to 10%. Simultaneously, the method also reported a loss in very few cases of a specific dataset, as indicated by the ↓ symbol in Table 5. Additionally, *DAPT* showed an average gain of more than 4% compared with *TAPT*, and up to approximately 9% as we see with *mBERT* with the *feature-based* approach. Overall, applying the *fine-tuning* approach over *DAPT* models proved to be the best pipeline among all the different versions of the language models considered in this study, from the original checkpoints to adapted models through TAPT and DAPT. This is demonstrated when we examine models individually and scores by dataset, as shown in Table 6.

Although we confirmed that continued pre-training strategies do confer essential additional knowledge to language models, as observed mainly with *DAPT*, experiments revealed that such enhancements are still suboptimal when compared to training from scratch, as our *BERTweet.BR* provides better results for the majority of the scenarios in this study, whether we apply the *feature-based* or *fine-tuning* approach. These findings address research question *RQ1*, as we have demonstrated the effectiveness of training a completely new model for tweets. Compared with the previous best results per method, *BERTweet.BR* delivered gains of up to 4.5% in *feature-based* and up to 4.2% in the *fine-tuning* approach.

In summary, the results presented in this section highlight the potential of models crafted for a specific language and domain as being more suited to handle social media tasks. In particular, *BERTweet.BR* fills the gap in the domain of Portuguese tweets, as it substantially outperforms the baselines in the vast majority of scenarios.

6 Qualitative analysis

Upon examining the results reported in this study, it is noteworthy that all studied models reported their highest and lowest performances in two particular datasets: *tweemg*, as the one with the top f1-scores, and *unilex* with nearly all the lowest scores, as depicted in Table 7. In fact, the evaluation of this table discloses a notable similarity in the performance rankings across the models, despite their distinct domains and pre-training procedures. For instance, besides *tweemg*, all models ranked *sentbr* and *comput*

Table 7 Performance ranking of benchmark models, listing in ascending order the datasets according to the weighted f1-score obtained by each model when applied *feature-based* and *fine-tuning* approaches on top of the *original checkpoints* of the models and

adapted versions resulting from *domain-adaptive pre-trained* (DAPT) procedure. Note that the top performances of all benchmark models were obtained in *tweemg* dataset, while the lowest performances of almost all models have registered in the *unilex* dataset

Origanl checkpoint	Feature- based	Fasttext		Bertimbau		Mbert		Xlmr		Xlmt		Bertweetbr	
		Unilex	0.661	Unilex	0.668	Unilex	0.602	Unilex	0.649	Unilex	0.672	Unilex	0.679
		<i>Covidbr</i>	0.770	<i>Mining</i>	0.784	<i>narrpt</i>	0.651	<i>narrpt</i>	0.738	<i>covidbr</i>	0.768	<i>covidbr</i>	0.778
		<i>Narrpt</i>	0.776	<i>Narrpt</i>	0.804	<i>mining</i>	0.706	<i>covidbr</i>	0.771	<i>mining</i>	0.802	<i>mining</i>	0.824
		<i>Mining</i>	0.776	<i>Covidbr</i>	0.805	<i>covidbr</i>	0.716	<i>mining</i>	0.777	<i>fiat</i>	0.826	<i>fiat</i>	0.827
		<i>Comput</i>	0.781	<i>Comput</i>	0.821	<i>sentbr</i>	0.738	<i>comput</i>	0.799	<i>comput</i>	0.834	<i>narrpt</i>	0.843
		<i>Fiat</i>	0.815	<i>fiat</i>	0.826	<i>comput</i>	0.752	<i>sentbr</i>	0.803	<i>narrpt</i>	0.838	<i>comput</i>	0.848
		<i>Sentbr</i>	0.828	<i>Sentbr</i>	0.842	<i>fiat</i>	0.774	<i>fiat</i>	0.807	<i>sentbr</i>	0.851	<i>sentbr</i>	0.881
		<i>Tweemg</i>	0.945	Tweemg	0.954	tweemg	0.927	tweemg	0.942	tweemg	0.949	tweemg	0.954
	Fine- tuning			Unilex	0.732	narrpt	0.688	unilex	0.715	unilex	0.723	unilex	0.733
				<i>Mining</i>	0.828	<i>unilex</i>	0.703	<i>mining</i>	0.732	<i>covidbr</i>	0.806	<i>covidbr</i>	0.828
				<i>Narrpt</i>	0.855	<i>mining</i>	0.759	<i>comput</i>	0.742	<i>mining</i>	0.829	<i>mining</i>	0.857
				<i>Comput</i>	0.855	<i>comput</i>	0.773	<i>narrpt</i>	0.742	<i>comput</i>	0.842	<i>comput</i>	0.874
				<i>Covidbr</i>	0.864	<i>covidbr</i>	0.786	<i>covidbr</i>	0.768	<i>narrpt</i>	0.873	<i>fiat</i>	0.892
				<i>Sentbr</i>	0.877	<i>sentbr</i>	0.841	<i>sentbr</i>	0.825	<i>sentbr</i>	0.882	<i>sentbr</i>	0.902
				<i>Fiat</i>	0.891	<i>fiat</i>	0.878	<i>fiat</i>	0.862	<i>fiat</i>	0.888	<i>narrpt</i>	0.909
				Tweemg	0.972	tweemg	0.972	tweemg	0.967	tweemg	0.971	tweemg	0.973
				Bertimbau		mbert		xlmr					
				Unilex	0.674	unilex	0.673	unilex	0.675				
				<i>Covidbr</i>	0.781	<i>covidbr</i>	0.773	<i>covidbr</i>	0.766				
				<i>Mining</i>	0.816	<i>mining</i>	0.809	<i>mining</i>	0.821				
				<i>Fiat</i>	0.830	<i>fiat</i>	0.821	<i>fiat</i>	0.824				
				<i>Comput</i>	0.834	<i>comput</i>	0.830	<i>comput</i>	0.831				
				<i>Narrpt</i>	0.840	<i>narrpt</i>	0.848	<i>narrpt</i>	0.861				
				<i>Sentbr</i>	0.863	<i>sentbr</i>	0.861	<i>sentbr</i>	0.867				
				Tweemg	0.955	tweemg	0.956	tweemg	0.961				
				Unilex	0.735	unilex	0.735	unilex	0.731				
DAPT	Feature- based			<i>Covidbr</i>	0.824	<i>covidbr</i>	0.826	<i>covidbr</i>	0.823				
				<i>Mining</i>	0.852	<i>mining</i>	0.848	<i>mining</i>	0.849				
				<i>Comput</i>	0.864	<i>comput</i>	0.866	<i>comput</i>	0.869				
				<i>Narrpt</i>	0.887	<i>narrpt</i>	0.882	<i>narrpt</i>	0.889				
				<i>Fiat</i>	0.889	<i>sentbr</i>	0.885	<i>fiat</i>	0.890				
				<i>Sentbr</i>	0.890	<i>fiat</i>	0.888	<i>Sentbr</i>	0.894				
				Tweemg	0.972	Tweemg	0.973	Tweemg	0.972				
	Fine- tuning												

The values in bold show the highest and lowest performances

datasets within the top four highest performances when following the *feature-based* approach. Likewise, the models also placed *mining* and *covidbr* datasets within their four lowest-performing lists, along with *unilex*. Moreover, *BERTweet.BR* and *XLM-T*, which are the sole models initially pre-trained for tweets, appear to concur on the degree of difficulty across datasets. They reported an identical

order in their rankings following *fine-tuning* and exhibited only one discrepancy in relation to *feature-based* approach. The other remaining models, pre-trained for generic domain, also share most of their rankings as they reported identical lists of the top four highest and bottom four performance independently of the approach. More precisely, when *fine-tuning* is applied they all positioned

covidbr, *sentbr*, *fiat* and *tweetmg* datasets as top performers in ascending order. Lastly, we highlight that generic-domain models display a higher degree of congruence after *domain-adaptive pre-training*. As illustrated by the DAPT rankings in the latter half of Table 7, the three adapted models now exhibit a perfect match across rankings when implementing the *feature-based* method and only a single difference when *fine-tuning* is applied. As previously generic models become adapted to tweets, their results increasingly resemble those of XLM-T and *BERTweet.BR*, which confirms the effectiveness of adaptation methods. As a result, these models now share the lists of the top four and bottom four performances with XLM-T and *BERTweet.BR*, although the exact order varies.

What we observe from the aforementioned findings is that despite the variation in weighted F1-scores across different models and datasets, there exists a discernible pattern of performance clusters, grouping datasets into levels of difficulty and their corresponding compatibility with models. This motivated the examination of two datasets of the clusters with the aim of clarifying potential factors accounting for the highest and lowest performances of the collection: *unillex* and *tweetmg*.

Unillex The *unillex* dataset presents the most significant challenge among the collection as evidenced by weighted f1-scores ranging between 0.602 and 0.735. While these scores may not be deemed low, they underscore that the *unillex* dataset does pose the most difficulty for all of the models.

First, the topic of politics is a subjective matter in which opinions diverge among individuals based on their personal experiences, values, and cultural backgrounds. This is particularly evident in recent contexts, marked by heightened polarization in political views, made even more intense by the popularity of social networks. In this regard, determining whether a tweet carries a positive or negative opinion out of this subjectivity is indeed a more difficult task. To illustrate, when we extract the list of most frequent words in positive and negative tweets of *unillex*, on average three-fifths of them are presented in both lists. Also, more than 40% of the most frequent words are present in all subsets of positive, negative, and neutral tweets. This evidences the subjectivity of the matter of politics and how challenging it is to spot the sentiment present in its tweets.

Particularly, when we look at the twenty most frequent words in tweets of *unillex* that are labeled as positive and negative as shown in Table 8, we see that words like *#brasil*, *#dilma*, *#lula*, *#pcdob*, *#pmdb*, *#psdb*, *#pt*¹¹, ¹²and *governo*¹³ exists in both subsets, emphasizing the absence

Table 8 The twenty most frequent words in *Unillex* dataset and their absolute frequency within each subset of tweets labeled as *positive*, *negative*, and *neutral* sentiments. In bold, words that appear in all three subsets (6). Underlined words appear in both *positive* and *negative* subsets. In italic, words that appear in both *positive* and *neutral* subsets. Double-underlined words appear in both *negative* and *neutral* subsets

Word	Pos	Neg	Neu
#pt	1521	2568	1354
#psdb	508	453	183
#brasil	238	350	416
#pmdb	171	388	412
#pcdob	154	268	488
<i>#psd</i>	137	–	368
<i>brasil</i>	183	256	–
<i>#pp</i>	128	–	339
<i>#pr</i>	253	–	256
#dilma	140	270	131
<u>governo</u>	136	269	–
<i>#novo</i>	129	–	361
<i>golpe</i>	–	270	–
<i>psd</i>	–	–	351
<i>#mst</i>	–	–	228
<i>#rede</i>	134	–	141
<i>#pdt</i>	–	–	230
<i>contra</i>	–	212	–
<u>#golpe</u>	–	176	131
<u>#lula</u>	106	192	–

of a discernible pattern to distinguish sentiment. Names of Brazilian politicians and political parties as well as words as *brasil* and *governo* — which are more likely to convey a neutral connotation — are consistently found in both positive and negative tweets. Additionally, when analyzing the twenty most frequent words in neutral tweets, an almost identical list of intersecting terms emerges: *#brasil*, *#dilma*, *#pcdob*, *#pmdb*, *#psdb*, *#pt* and *governo*, thereby reaffirming the lack of a pattern for identifying sentiment as the same words are frequent in tweets with three different sentiments. The absence of *#lula*¹⁴ in this neutral list of frequent words implies that the name of the former president is associated with either positive or negative sentiment, but not neutral, a fact that once more exemplifies the politically polarized climate in Brazil.

In fact, a comprehensive examination of the dataset vocabularies and their intersections, as illustrated in Table 9, demonstrates that *covidbr*, *mining* and *fiat* compose the group of datasets that exhibit the highest degree of similarity relative to *unillex* among the entire collection. Remarkably, these four datasets correspond to the same set

¹¹ PT, PSDB, PMDB and PCdoB are Brazilian political parties. https://en.wikipedia.org/wiki/List_of_political_parties_in_Brazil., brasil, dilma

¹² Dilma Rouseff, politician who served as the 36th president of Brazil, holding the position from 2011 until her impeachment and removal from office on 31 August 2016. https://en.wikipedia.org/wiki/Dilma_Rouseff..

¹³ *government* in Portuguese.

¹⁴ Luis Inacio Lula da Silva, also known as Lula da Silva or simply Lula, is the 39th and current president of Brazil. https://en.wikipedia.org/wiki/Luiz_In%C3%A1cio_Lula_da_Silva..

Table 9 Vocabulary similarity matrix representing the lexical congruence among various datasets, considering the top 1,000 most frequent words within each dataset, after excluding stopwords present in the NLTK Portuguese dictionary. The degree of similarity is

	Covidbr	Sentbr	Fiat	Narrpt	Mining	Compbr	Tweemg	Unilex
Covidbr	x	28.6	31.0	25.3	33.1	<u>24.9</u>	28.3	42.8
Sentbr	28.6	x	38.9	35.9	39.2	33.5	<u>17.6</u>	39.7
Fiat	31.0	38.9	x	36.7	41.2	38.7	25.4	42.4
Narrpt	25.3	35.9	36.7	x	37.9	38.0	<u>18.8</u>	35.0
Mining	33.1	39.2	41.2	37.9	x	42.3	<u>24.1</u>	43.0
Compbr	<u>24.9</u>	33.5	38.7	38.0	42.3	x	<u>17.8</u>	33.4
Tweemg	28.3	<u>17.6</u>	25.4	<u>18.8</u>	<u>24.1</u>	<u>17.8</u>	x	32.1
Unilex	42.8	39.7	42.4	35.0	43.0	33.4	32.1	x

quantified by the proportion of shared terms present in both vocabularies with a maximum of 100. The five highest similarities are emphasized in bold, while the five lowest percentage similarities are presented underlined

of the four most challenging datasets for the Twitter-domain models, as shown in Table 7. This finding implies that vocabularies do play a crucial role in influencing the performance of models in the sentiment classification tasks, as corroborated by the *feature-based* scores of Twitter models in the form of original checkpoints (*BERTweet.BR* and *XLM-T*) or after adaptation of mBERT, BERTimbau and XLM-R.

To conclude, an additional challenging aspect of *unilex* is the high presence of very short tweets with approximately 25% of its messages comprising fewer than ten words. Although a substantial proportion of tweets of *unilex* does contain hashtags (90%), there is a relatively low incidence of user mentions (about 20% of tweets) and minimal inclusion of emoticons and emojis (less than 10%). The presence of URLs and email addresses is virtually absent. Given that tweets are typically characterized by concise sentences that frequently rely on user mentions, hashtags, and emoticons, it can be suggested that *unilex* does not accurately adhere to a conventional *tweet-like* dataset, resulting in decreased compatibility with models specifically designed for the tweets domain. As a result, this gap makes applying pre-trained Twitter models like *BERTweet.BR* and *XLM-T* more challenging. We also observed that this difficulty is also applied to the Twitter-adapted versions of models such as mBERT, BERTimbau, and XLM-R.

TweetsMG The *tweemg* dataset emerges as the best-performing dataset for all models analyzed in this study, achieving a minimum score of 0.927. It boasts the highest number of retweets (29.6%) by a significant margin and presents the highest proportion of emoticons (83.92%), followed by *covidbr* (70.83%). Additionally, it has the highest proportion of URLs (82.23%), with *covidbr* trailing at 69.00%. Regarding user mentions, *tweemg* holds the second place with a proportion of 49.20%, only surpassed

by *mining* at 68.29%. The attributes above illustrate a notable distinction between *tweemg* and *unilex* with regard to their adherence to *tweet-like* datasets and the impact of it on their average scores. What we observed is that the degree of alignment between a dataset and the *standard tweet* positively impacts the performance of the model, as evidenced by the strong results obtained with *tweemg* in contrast to the relatively low average scores of *unilex*.

Tweemg integrates intrinsic features that facilitate the clustering of its tweets, thereby simplifying the text classification process. First, it follows a balanced distribution in terms of classes. Then, we observed that, on average, positive tweets in *tweemg* tend to exhibit shorter lengths than negative ones. Furthermore, in contrast to *unilex*, vocabularies of *tweemg* display low intersection rates (averaging less than 15%) when evaluating the most frequent words within the groups of positive, negative, and neutral tweets. Mainly, when analyzing the twenty most common words in *tweemg* by sentiment, only three appear concurrently in both positive and negative subsets. This suggests the presence of virtually independent vocabularies for each sentiment category, unlike what we observe with *unilex*, where we encounter three times as many the number of shared words among all distinct groups. This indicates the lack of a discernible pattern to differentiate classes in the most challenging dataset in the collection, positioning *unilex* and *tweemg* in opposite directions. Objectively, the relative independence of vocabularies of the *tweemg* dataset results in reduced entropy, which subsequently facilitates the job of machine learning algorithms in splitting tweets into groups of different sentiments.

This specific ability of *tweemg* is further evidenced when we plot the embeddings extracted from pre-trained models after dimensionality reduction using t-SNE [50] as shown in Fig. 3. In the presented figures, distinct colors represent different labels: red corresponds to negative

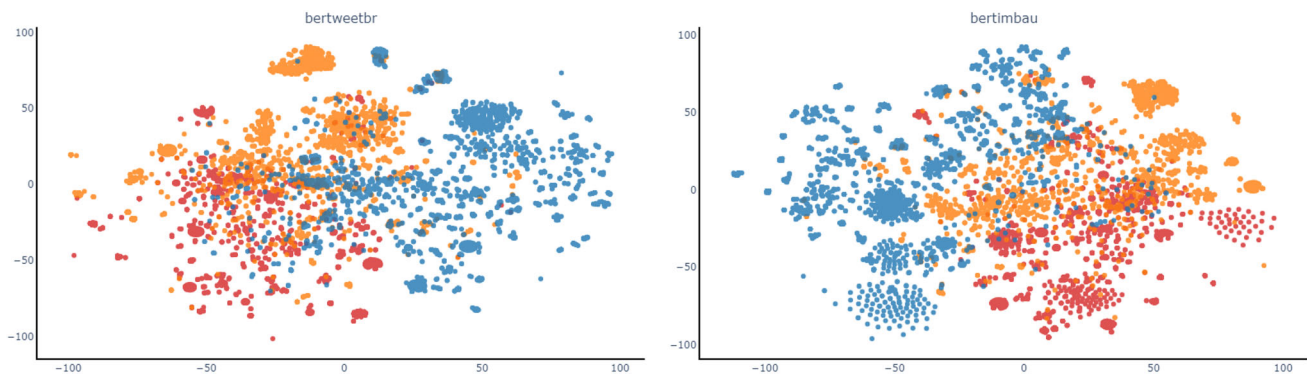


Fig. 3 Visualization of embeddings for the *tweemg* dataset. Each point corresponds to an individual tweet in the dataset, with 768-dimension embeddings derived from pre-trained models BERTweet.BR and BERTimbau. Dimensionality reduction was performed using t-SNE. The color scheme denotes the three sentiment labels: red

for negative sentiment, blue for positive sentiment, and orange for neutral sentiment. Distinct sentiment clusters are discernible, highlighting the similarity among tweets sharing the same label and potentially indicating the ease of detecting sentiments within the *tweemg* dataset

sentiment, blue to positive sentiment, and orange to neutral sentiment. Embeddings generated by *BERTweet.BR* and *BERTimbau* suggest that groups of tweets with the same labels in the *tweemg* dataset also possess similar semantic meanings, as illustrated by the proximity of their vector representations in the two-dimensional space. Groups of tweets with the same label appear clustered together while distant from tweets of other sentiments. Consequently, it becomes easier to infer the sentiment of a given tweet visually. In fact, this is an intrinsic characteristic of this dataset as, regardless of the pre-trained model used to generate the embeddings, tweets from the *tweemg* dataset prove to be reasonably distinguishable when visualized in the plots. Precisely, samples from the *tweemg* dataset are more readily identifiable using t-SNE technique compared to all other datasets in the collection. For instance, samples with different labels in *unilex* dataset are virtually indistinguishable as depicted in Fig. 4.

Effect of specific Tokenizer Although training a specific vocabulary on tweets cannot guarantee the absence of *out-of-vocabulary* tokens, it does provide a more concise way of representing sentences. This is demonstrated in Table 10, which presents the statistics of each dataset after normalization and tokenization. For each dataset is indicated the minimum, maximum, and average length in tokens after each model tokenizer. In *out-of-vocabulary* (OOV), it is provided the number of rows, the total number of occurrences, and the quantity of unique OOV found in each dataset and after each particular tokenizer (*#has*, *#total* and *#unique*, respectively). The *unique max* column indicates the maximum number of occurrences of one OOV token in the same tweet. Finally, it is presented the three most frequent *out-of-vocabulary* tokens along with their frequency in each of the datasets. Note that for *XLM-R* and *XLM-T*, models with the largest vocabularies, tokenizers were able to decode the whole dataset. On average, *BERTweet.BR* tokenizer generates the shortest decoded

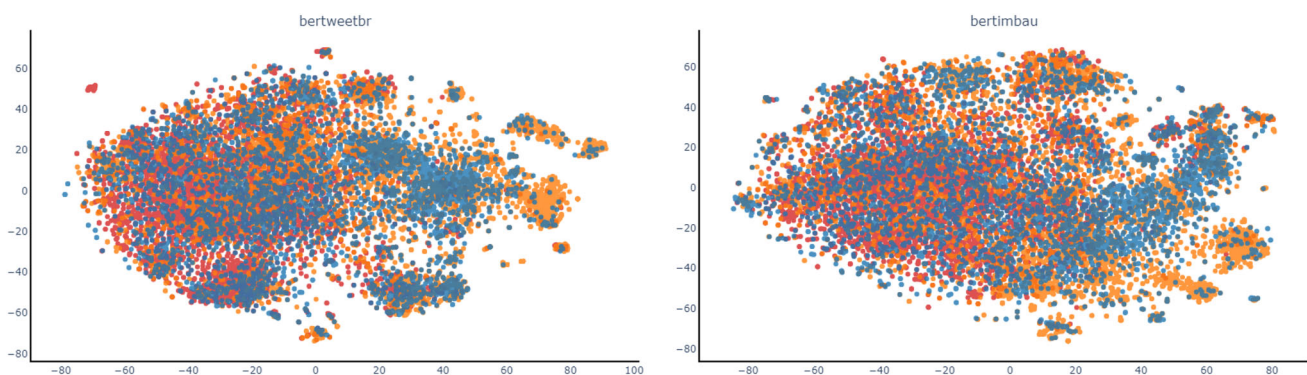


Fig. 4 Visualization of embeddings for the *unilex* dataset. Each point corresponds to an individual tweet in the dataset, with 768-dimension embeddings derived from pre-trained models BERTweet.BR and BERTimbau. Dimensionality reduction was performed using t-SNE. The color scheme denotes the three sentiment labels: red for negative

sentiment, blue for positive sentiment, and orange for neutral sentiment. In contrast to the *tweemg* dataset, the plot visually illustrates the inherent challenge in discerning sentiment within this dataset, as it is virtually impossible to identify a clear semantic separation among the different labels of the tweets

Table 10 Statistics of each dataset after normalization and tokenization. (*) Based on classical TwitterTokenizer from NLTK package.

		Tokens			Out-of-Vocabulary				Most freq.
		Min	Avg	Max	#Has	#Total	#Unique	Unique max	
Covidbr	Fasttext*	6	32.9	68	514	1456	484	22	covid (75); #coronavárus (57); #coronavirus (45)
	Bertimbau	13	52.6	138	31	41	16	2	q (22); 2ª (3); 25 (2)
	Mbert	14	54.2	146	31	55	10	2	“(20);”(19);‘(6)
	Xlmr	11	52.6	164	0	0	0	0	–
	Xlmt	11	52.6	164	0	0	0	0	–
	BERTweet.BR	8	39.9	85	18	49	19	2	🙏 (8); antônio (2); 🙌 (2)
Sentbr	Fasttext*	2	13.6	47	7769	12532	1354	11	#masterchefbr (2761); #encontro (1091)
	Bertimbau	6	24.3	82	454	568	43	6	q (424); ñ (33); ♥ (28)
	Mbert	7	24.8	97	8	15	11	6	”(2); ‘ (2);–(1)
	Xlmr	7	24.0	124	0	0	0	0	–
	Xlmt	7	24.0	124	0	0	0	0	–
	BERTweet.BR	4	18.1	56	29	44	20	6	😊 (11); \u200d ♀ (5); 😞 (4)
fiat	Fasttext*	2	19.1	43	6946	11831	2295	11	#fiat (140); #cartolaafc (116); 1507 (71)
	Bertimbau	7	30.9	111	387	471	72	3	q (280); ¬ (34); ñ (16)
	Mbert	7	30.4	72	5	5	3	1	‘(3); ¢(1);’(1)
	Xlmr	6	29.3	70	0	0	0	0	–
	Xlmt	6	29.3	70	0	0	0	0	–
	BERTweet.BR	4	24.3	63	244	247	13	2	\x97 (226);\x99 (3); ariquemes (3)
Narprt	Fasttext*	2	16.6	38	504	929	353	6	:d (13); #sorteio (12); #promoção (10)
	Bertimbau	5	27.5	59	53	69	17	2	q (33); ¬ (11); qe (16)
	Mbert	6	27.4	54	16	22	7	2	“(8);”(6);–(4)
	Xlmr	4	24.9	50	0	0	0	0	–
	Xlmt	4	24.9	50	0	0	0	0	–
	BERTweet.BR	4	20.7	43	0	0	0	0	–
Mining	Fasttext*	2	16.9	44	2018	4846	753	12	#claro (104); #bradesco (91); #vivoemrede (85)
	Bertimbau	6	28.5	83	156	196	19	2	q (138); ¬ (14); ñ (10)
	Mbert	7	29.4	56	0	0	0	0	–
	Xlmr	6	28.2	53	0	0	0	0	–
	Xlmt	6	28.2	53	0	0	0	0	–
	BERTweet.BR	4	21.3	46	64	64	5	1	#bancodobrasil (57); #bancocentralbr (2)
compbr	Fasttext*	1	19.1	55	1647	3197	673	11	#dell (75); #notebook (32); v14t (22)
	Bertimbau	5	33.1	89	107	134	27	3	q (81); 5ª (7); ñ (6)
	Mbert	6	31.0	95	28	33	7	2	- (13); ”(9); —(4)
	Xlmr	4	27.9	92	0	0	0	0	–
	Xlmt	4	27.9	92	0	0	0	0	–
	BERTweet.BR	4	24.9	72	4	4	4	1	😊 (1); \x99 (1); aspect (1)
tweemg	Fasttext*	1	18.2	38	8038	16885	847	10	#timbeta (206); #globo (130); #operacaobetalab (121)
	Bertimbau	6	31.2	62	471	1100	49	3	q (676); ñ (312); 16ª (15)
	Mbert	5	31.1	58	101	121	6	2	–(37);‘(28); ”(19)
	Xlmr	4	29.2	62	0	0	0	0	–
	Xlmt	4	29.2	62	0	0	0	0	–
	BERTweet.BR	3	22.3	51	102	107	11	2	#raynniere (75); timóteo (13); 😊 (4)
unilex	Fasttext*	1	17.2	73	11632	43693	9531	27	#pt (5444); #psdb (1144); #brasil (1004)
	Bertimbau	3	32.3	176	792	1123	147	7	q (469); ñ (105); • (47)
	Mbert	4	33.3	202	288	483	51	7	“(147);”, (133);–(43)

Table 10 (continued)

	Tokens			Out-of-Vocabulary				
Xlmr	3	32.4	205	62	179	80	10	😄 (13); ❤️ (12); 😡 (8)
Xlmt	3	32.4	205	62	179	80	10	😄 (13); ❤️ (12); 😡 (8)
BERTweet.BR	3	24.3	72	480	1208	169	7	👏 (85); #dilmrousseff (79); paschoal (38)

sentences across all datasets, demonstrating that our model's tokenizer has successfully learned highly specific tokens for a corpus of Portuguese tweets, enabling it to generate concise representations. Comparatively, using the plain Portuguese tokenizer of BERTimbau on the of the same datasets resulted in decoded sentences 33% longer.

The XLM-R and XLM-T multilingual tokenizers possess the largest vocabulary among all models, encompassing 250,002 entries. This is over double the size of the second-largest vocabulary, which contains 119,547 entries and belongs to the mBERT model. As expected, the vocabulary size influences the number of *out-of-vocabulary* (*OOV*) instances. The only dataset in which XLM-R and XLM-T tokenizers exhibit *OOV* occurrences is *unilx*. It can be expected that a low frequency of *OOV* tokens may contribute to improved results, as evidenced by the XLM-T model, which has virtually no *OOV* tokens across all datasets. However, it is also observed that models such as BERTimbau, which has the smallest vocabulary size and the highest frequency of *OOV* in all datasets, as shown in Table 10, recorded significant results trailing only behind *BERTweet.BR* in several instances.

7 Conclusion

This work presented the first public large-scale pre-trained model specific to the Portuguese tweet domain. To assess our model, we explored several pipelines, applying both *feature-based* and *fine-tuning* approaches on top of *off-the-shelf* language models as well as adjusted models to the context of Twitter in Portuguese. We found that *BERTweet.BR*, trained from scratch, performed better than its competitors in most scenarios of sentiment classification with datasets of varying sizes. *BERTweet.BR* achieved the absolute best F1-score on 36 out of 48 experiments while matching the previous best performances on two scenarios, which certifies the effectiveness of a domain-specific language model pre-trained for Portuguese tweets. This provides a positive response to inquiries about research questions *RQ1* and *RQ2*.

In this regard, our work proved to be a relevant step forward in the field of natural language processing in Portuguese, which we release model and code from the transformers library¹⁵ and Github¹⁶ to advance future research in analytical tasks for Portuguese.

8 Future work

Due to the dynamic nature of debates on the Twitter platform, one natural next step is to incorporate new knowledge into our model with the latest tweets from 2020 onward. In particular, our findings showed that *BERTweet.BR* did poorly with the *opcovidbr* dataset. Acquiring data after that point and retraining the models could lead to different results. We also would like to investigate *BERTweet.BR* performance on other classification and sequence-based tweets tasks.

Finally, in light of the significant recent advancements in NLP related to large language models (LLMs) like GPT-4 [36], LLaMA [49], and BLOOM [56], next we plan to follow in-context learning applying the few/one/zero-shot approaches [6] to LLMs and compare these performances to *BERTweet.BR*, a scenario we did not explore in this study.

8.1 Availability of data and materials

Despite Portuguese being among the top ten languages utilized on the Internet as of January 2020,¹⁷ there remains a scarcity of resources for sentiment analysis in this language. Motivated by the necessity to assemble a comprehensive Portuguese corpus, we explored the literature for annotated resources pertaining to sentiment analysis and successfully acquired eight datasets by contacting authors. These datasets, comprising human-annotated tweets,

¹⁵ <https://huggingface.co/melll-uff/bertweetbr>.

¹⁶ <https://github.com/MeLLL-UFF/BERTweet.br>.

¹⁷ <https://www.statista.com/statistics/262946/share-of-the-most-common-languages-on-the-internet/>.

exhibit diversity in subject matter, with three being binary and five being multiclass. The labeled datasets can be accessed at <https://bityli.com/RvhFax..>, and we have also made the collection available in the transformers datasets library.¹⁸

OPCovid-BR The OPCovid-BR dataset consists of 600 manually labeled tweets about the Covid-19 pandemic posted by Brazilian Twitter users. This is the smallest dataset in our collection but also the one with the longest tweets (approximately 28 words on average, almost double the second dataset in this regard). The authors developed a Twitter API to extract tweets using the key term search: “coronavirus”. The OPCovid-BR were annotated by three annotators, with concordance among annotators equal to 82,77%. It is annotated with the binary document polarity (positive and negative) and fine-grained opinion (explicit aspects) for each tweet. There are 300 tweets with positive labels and 300 tweets with negative labels. For simplicity, in this paper, we refer to this dataset as *covidbr*. This dataset is available in the transformers datasets library at <https://huggingface.co/datasets/melll-uff/opcovidbr..>

TweetSentBR The dataset TweetSentBR was created in 2017 by Brum and das Graças Volpe Nunes [7] and is a corpus of tweets in Brazilian Portuguese in the domain of TV shows. This is a reasonably unbalanced dataset that contains the shortest tweets on average (approximately 11 words in each), and more than 40% of the rows have less than 10 words. Virtually all tweets have at least one occurrence of hashtags. TweetSentBR is also the dataset with the highest frequency of emojis (22.56%). The dataset was labeled by several annotators following steps established in the literature to improve the reliability of the Sentiment Analysis task and each tweet was annotated as either positive or negative. Several annotators labeled each tweet as positive or negative following an annotation process of eight steps. The final label for each tweet was determined based on a major voting strategy. While some tweets were labeled by only one annotator, others were annotated by three or seven. For simplicity, we refer to this dataset as *sentbr* in this manuscript. This dataset is available in the transformers datasets library at <https://huggingface.co/datasets/melll-uff/tweetsentbr..>

FIAT-UFMG FIAT-UFMG is composed of tweets related to the “FIAT” brand, manually labeled as positive or negative. This is the second-largest dataset in our collection. Tweets were extracted by Martins et al [30] using the Twitter API filtering the year 2012 and messages related to the “Fiat” brand. For simplicity, we refer to this dataset as *fiat* in this study. This dataset is available in the transformers datasets library at <https://huggingface.co/datasets/melll-uff/fiat-ufmg..>

narr-PT The dataset narr-PT contains tweets that have been human-annotated with sentiment labels by three Mechanical Turk workers with the aim to create a multilingual sentiment dataset for the languages English, German, French, and Portuguese. Originally, in this dataset, there are 12,597 positive, neutral, and negative tweets. Particularly for Portuguese, there are 772 tweets. This balanced dataset is the smallest one in the collection we used in this study among the three-classe datasets. For simplicity, we refer to this dataset as *narrpt* in this manuscript. This dataset is available in the transformers datasets library at <https://huggingface.co/datasets/melll-uff/narrpt..>

MiningBR This dataset contains tweets of the companies with the most number of complaints in the Brazilian Consumer Protection and Defense Program agency (PROCON). At least two annotators manually labeled the tweets from this collection as neutral, positive, or negative according to their sentiment polarity. Because of the nature of its source, this is a highly unbalanced dataset with nearly 65% of the tweets belonging to the negative class and only 9% being positive. For simplicity, we refer to this dataset as *mining* in this manuscript. This dataset is available in the transformers datasets library at <https://huggingface.co/datasets/melll-uff/miningbr..>

Computer-BR This corpus consists of 2,281 tweets extracted in the period from January to September 2015. To build it, Moraes et al [32] used keywords related to computers, such as notebook, analysis, and testing, among others. In the annotation process, four human annotators have defined the polarity of the tweets, three of them participating in the whole process and the fourth deciding the final polarity in cases of disagreement only. It is worth mentioning that three annotators were from the Computer Science area and one of them was from the Linguistics area. To reduce noise, the dataset was normalized prior to publication: they removed (or treated) special characters and hashtags, turned emoticons and hyperlinks into text, and replaced abbreviations and slang with usual expressions, such as “vc” into “voc”(you) and “novis” into “novidade” (news). This dataset is also highly unbalanced, having its dominant class *neutral* with approximately 75% of all tweets. For simplicity, we refer to this dataset as *comput* in this manuscript. This dataset is available in the transformers datasets library at <https://huggingface.co/datasets/melll-uff/computerbr..>

TweetsMG TweetsMG is a multiclass dataset collected and labeled by the IT staff of Prodemge MG. It belongs to the domain of Education and Politics in the context of the Brazilian State of Minas Gerais. This dataset contains lots of duplicated tweets (retweets). For simplicity, we refer to this dataset as *tweemg* in this manuscript. This dataset is

¹⁸ <https://huggingface.co/melll-uff/..>

available in transformers datasets library at <https://huggingface.co/datasets/melll-uff/tweetsmg..>

UniLex The dataset UniLex, created by de Souza et al [47], contains most of the tweets belonging to the domain of politics. Labeling was performed by four annotators, each one labeling approximately 3,500 tweets. Tweets with dates, user mentions, and hashtags were considered neutral. This is the largest dataset of the collection used in this study. For simplicity, we refer to this dataset as *unillex* in this manuscript. This dataset is available in the transformers datasets library at <https://huggingface.co/datasets/melll-uff/unillex..>

Funding This research was financed by CNPq (National Council for Scientific and Technological Development), grants 307088/2023-5 and 315750/2021-9, FAPERJ- *Fundação Carlos Chagas Filho de Amparo à Pesquisa do Estado do Rio de Janeiro*, processes SEI-260003/002930/2024, SEI-260003/000614/2023, E-26/201.139/2022, and Coordenação de Aperfeiçoamento de Pessoal de Nível Superior-Brasil (CAPES)-Finance Code 001. We sincerely thank the reviewers for their thorough work and express our gratitude to all the researchers who made their datasets and code publicly available, enabling us to build upon their contributions.

Code availability The *BERTweet.BR* code, experimental results, and related documentation are publicly available on Github <https://github.com/MeLLL-UFF/BERTweet.br>.

Declarations

Conflict of interest The authors have no conflict of interest to disclose.

Datasets All datasets considered in this manuscript were gathered from previous work that made them publicly available. Although we have not directly collected any tweets, we are aware that using data collected from the Twitter platform should raise ethical reflections. Even though Twitter users assume their posts are not private, they are usually not explicitly informed that what they write can be used for scientific – our case – or commercial – not our case – purposes. Besides, they might usually assume that their tweets are ephemeral while they, in fact, can be collected and stored by anyone anywhere. We tried our best not to include sensitive content in our examples and not disclose the identity of their authors.

Language model Given that this work strongly relies on large-scale language models and datasets composed of social media texts, despite the best intentions, we anticipate possible ethical and social risks by perpetuating social biases and providing false or misleading information. In the case of language models, these risks usually spring from the chosen training *corpora* used to pre-train such large models. If your intent is to use our pre-trained model or a fine-tuned version in production, please be aware that, while *BERTweet.BR* like many other models is a powerful tool, it comes with limitations. To enable pre-training on large amounts of data, we scrape all the content we could find from Twitter until the year 2020, taking the best as well as the worst of what was available on this social media.

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